D-Flow / QGF
Overnight investigation

Third campaign | November 6-7, 2026
LAYOUT DRAFT - RESULTS STILL IN PROGRESS

Preserved40 evidence cutoff: 2026-11-07T08:11:35.235744+00:00
Main target100 evidence cutoff: 2026-11-07T08:28:14.001810+00:00

The current comparison uses four D-Flow families: regular or stochastic, each with raw or nearby-action averaged Q. Each family has a separate λ0 and λ1 condition, giving eight D-Flow conditions. All use identity-Jacobian guidance and complete eight-slice flow endpoints.

The controls are BC, original QGF at fixed κ=.02/.04, independently tuned original QGF, one original IPG and one original FK corrector. The latest main target is 100 heldout episodes per unique frozen configuration and trained model, with training seeds 11, 23 and 37. It may remain partial at the deadline. Older separately declared 40-episode panels remain 40.

Closed heldout evidence in this snapshot	Episode rows
New physical episodes, each CSV counted once	2560
Inherited complete A20 reference episodes	200
Training updates in this campaign	0

These completed-evidence counts deduplicate main A/B references across the 40 and 100 snapshots. They exclude ongoing validation, offline numerical checks and unfinished physical rollout cost. Pending cells are not zeros. Same-start comparisons are paired observations; 100 starts do not create 100 independently trained models.

This report supplements the preserved earlier campaigns. Historical synthetic points, custom navigation, coverage-stress navigation and cube manipulation must not be pooled with the standard native episodes here.

What does the agent actually do?

Tasks 1 and 5 use native MuJoCo PointMaze navigation with real state transitions. A point agent selects two-dimensional control actions to reach the task goal while navigating the maze. This is navigation, separate from the earlier cube-manipulation study.

Task 1 navigates toward nominal goal (20,20). Task 5 starts near nominal (20,4) and navigates toward (0,0). Resets may vary within the task-defined start region. The standard evaluation cap is 1000 control decisions, separate from the earlier modified 128-step coverage test.

BC and IQL have different jobs. BC learns the offline action distribution. IQL learns Q(s,a) and V(s) from offline transitions including next states. This campaign freezes the BC actor from update 1000 and the IQL critic from update 50000; samplers propose actions but do not retrain Q or V.

The guidance score is the minimum of the two frozen target-Q heads. It is an estimate of future task return, not a certificate that a proposed action is safe or well supported by the offline data. “Unseen Q” here means evaluating this learned critic on actions outside or poorly covered by that data.

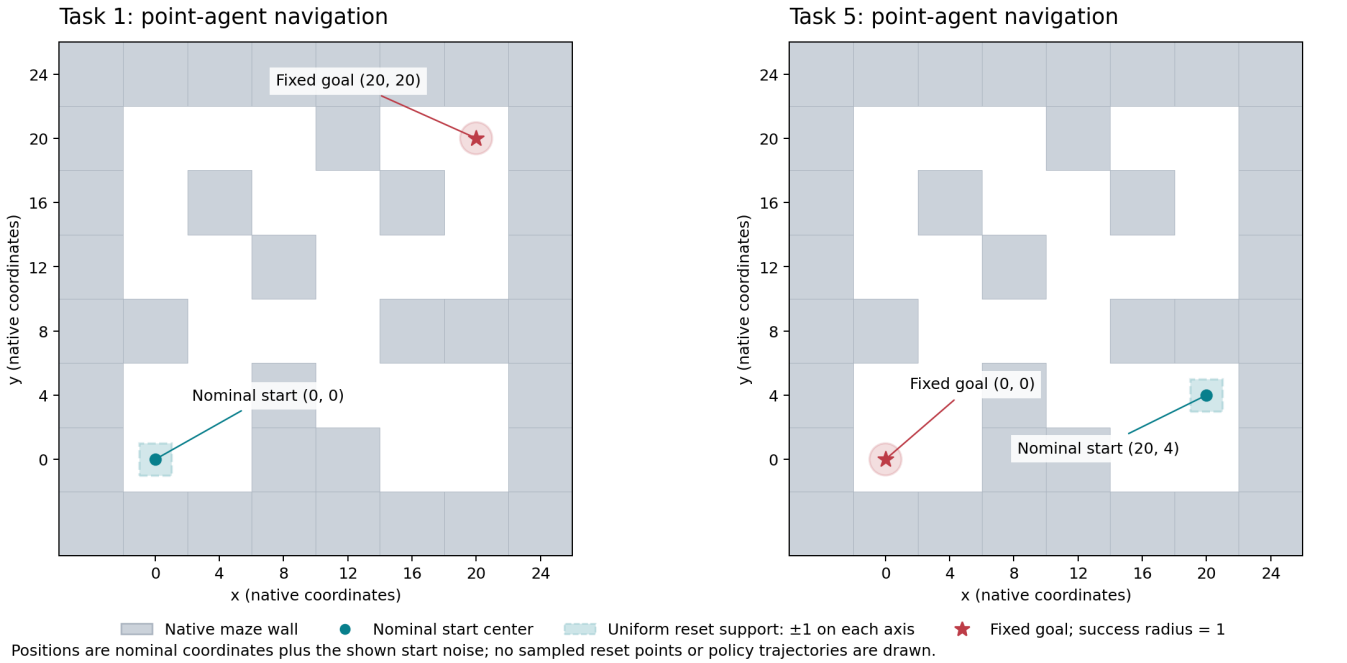
The native sparse reward is success minus one: -1 outside the radius-1 goal region, 0 on success. Success is checked before the action. The logged current position determines the relabeled reward and mask; IQL uses y=r+.99(1-success)V(actual next state), with V expectile .8. Larger Q predicts better discounted task return. The BC flow itself is ordinary behavior matching, with position input and no goal vector.

Seeds 11/23/37 identify independently trained actor/critic models. They are not task numbers or evaluation reset IDs. All three models remain in the report, including weak or failed ones.

Official maze geometry and task goals

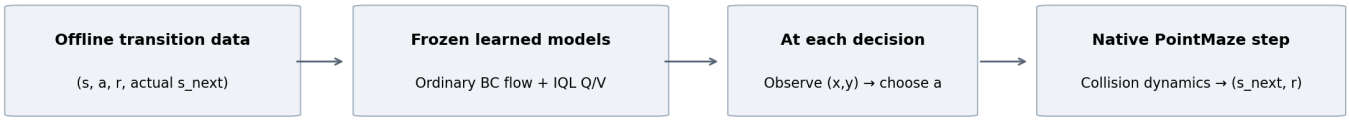
What the native experiments ask the agent to do

Navigate the point-agent around the actual walls to its fixed goal. Same official medium maze; different start and task reward.



Positions are nominal coordinates plus the shown start noise; no sampled reset points or policy trajectories are drawn.

Native task budget: up to 1,000 control decisions. Success is checked before the action; reward is −1 before success and 0 at success.



BC proposes actions; the fixed IQL-trained critic guides them. Q estimates discounted task reward and can be wrong; it is not an oracle.

These are sequential point-agent experiments with actual next states. The actor has 1,000 BC updates; the task-specific Q/V has 50,000 IQL updates.

Separate historical studies used a 128-decision TRAIN-anchor coverage stress or cube manipulation. Those resets, budgets and robot objectives differ.

Source-only drawing: official map/task literals and coordinate transform verified; no environment creation, reset, model call, physics or random draw.

Both tasks use this same official medium maze. Task1 starts around (0,0) and aims for (20,20); Task5 starts around (20,4) and aims for (0,0). Shaded start boxes show the native ±1 positional reset support, not sampled episodes. Red stars and radius-1 disks mark fixed goals. No policy route is drawn.

The figure comes from source-verified maze/task definitions and the exact coordinate transform. The model input is position (x,y); next states come from native collision dynamics. The flow time slices, D-Flow source iterations and physical control decisions are different time axes.

What κ, η, λ and averaging mean

Let F8(s,z) be the complete eight-slice frozen BC decoder including final action clipping. Raw S(s,a)=min(Q1,Q2). Averaged S is the mean over K nearby clipped actions of min(Q1,Q2); the minimum is taken before averaging.

Operation	Meaning in this campaign
κ (kappa)	Multiplier on Q-gradient guidance. Independently selected by task, trained model, method and fixed λ.
η (eta)	Step size for each D-Flow source update.
λ (lambda)	Coefficient of source-prior pull -λz. λ0 removes this pull; λ1 includes it.
I	Number of outer D-Flow source updates. Each update uses a full-flow endpoint.
σ / K	Nearby-action perturbation scale and total averaging samples. The antithetic bank stays fixed throughout one action generation.

Identity guidance: g = grad_a S(s,a), evaluated at a=F8(s,z), without multiplying by the endpoint Jacobian. Regular update: z ← z + η(κg - λz). Stochastic update adds sqrt(2ηT) ξ, with T=1. The executed action is F8(s,z_I).

Setting κ=0 removes Q guidance. It does not necessarily reproduce BC: λ1 still pulls z toward zero, and stochastic source noise still moves z. The unchanged BC flow remains a separate canonical control.

Source noise moves z. Q averaging changes the action score. Neither operation establishes correction of broad critic extrapolation error. Because the identity field is a surrogate, these runs do not imply exact Langevin sampling of Q(F8(z)). MH is outside this campaign’s scope.

Full-flow endpoints integrate all eight BC time slices. The excluded one-step D-Flow endpoint extrapolation is different from the numerical integrator used inside this complete flow. Original QGF still refreshes its own velocity correction and endpoint estimate at every flow time slice. QGF has no D-Flow η, λ or outer I setting.

Original QGF at each t=i/8: v_i=v(s,a_i,t); e_i=clip(a_i+(1-t)v_i); g_i=grad_a Q(s,a) evaluated at e_i; a_(i+1)=a_i+(v_i+κg_i)/8. It recomputes v_i, e_i and g_i at all eight slices, then applies its final action clipping. Its κ multiplies a velocity correction; D-Flow κ acts inside the repeated source update, so equal numerical κ values need not mean equal action changes.

Independent selection domain: κ=0,.02,.04,.06,.08,.1,.2,.3,.4,.6,.8,1,2,3,5,8,12,20,40,80; η=.005,.01,.03,.1; I=1,2,4,8,16,32,64,128,256; σ=.01,.03,.1,.2; K=4,16. Averaged-Q search uses 720 base settings plus at most 70 refinements per condition, followed by 8-episode physical validation. This hierarchy is not an exhaustive σ/K Cartesian sweep.

Frozen configurations: standard native results

Task 1: original A20 + fresh B20

Method	Seed 11	Seed 23	Seed 37
BC flow	0/40	Pending	Pending
QGF fixed .02	0/40	Pending	Pending
QGF fixed .04	0/40	Pending	Pending
QGF tuned	29/40	Pending	Pending
IPG	22/40	Pending	Pending
FK corrector	19/40	Pending	Pending
DF Regular, raw Q, λ0	28/40	Pending	Pending
DF Regular, raw Q, λ1	32/40	Pending	Pending
DF Stochastic, raw Q, λ0	24/40	Pending	Pending
DF Stochastic, raw Q, λ1	24/40	Pending	Pending

Task 5: new A20 + new B20

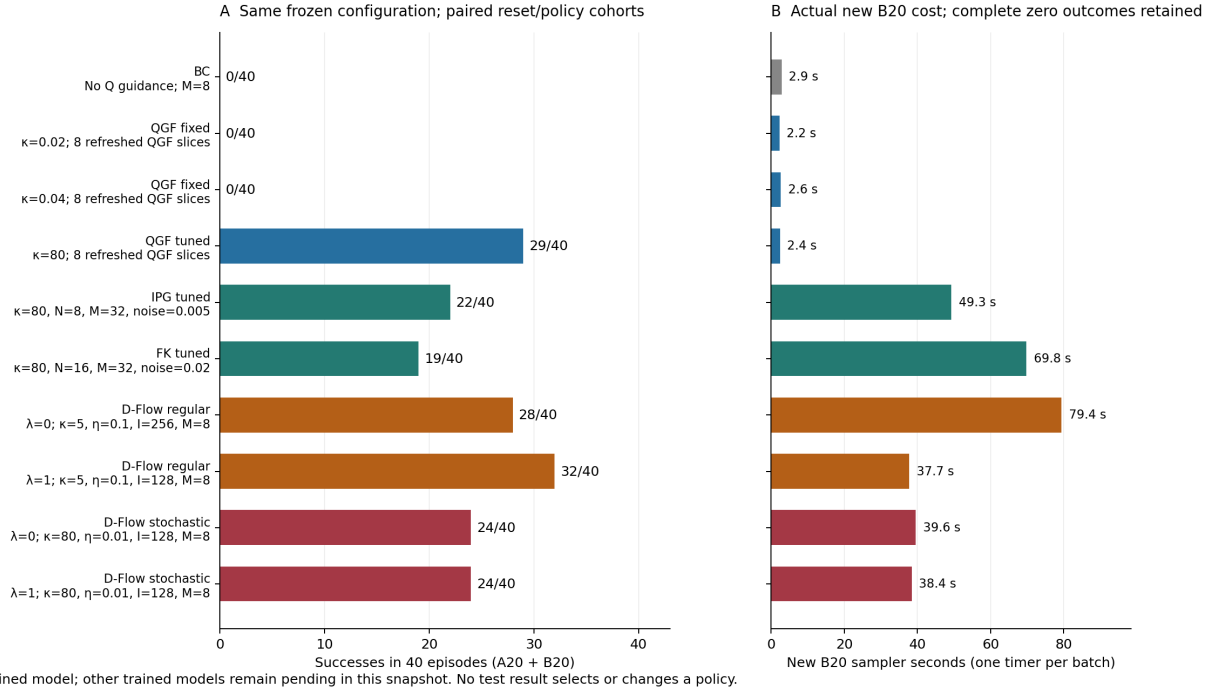
Method	Seed 11	Seed 23	Seed 37
BC flow	0/40	Pending	Pending
QGF fixed .02	0/40	Pending	Pending
QGF fixed .04	0/40	Pending	Pending
QGF tuned	37/40	Pending	Pending
IPG	40/40	Pending	Pending
FK corrector	40/40	Pending	Pending
DF Regular, raw Q, λ0	37/40	Pending	Pending
DF Regular, raw Q, λ1	40/40	Pending	Pending
DF Stochastic, raw Q, λ0	35/40	Pending	Pending
DF Stochastic, raw Q, λ1	40/40	Pending	Pending

Task1 may reference eligible original A20 and add fresh B20 at unchanged frozen settings. Task5 has its own separately declared new panel. All relevant three-model selections freeze before that branch's tests. Reported totals require both disjoint cohorts and an identical selected configuration.

First completed model: success and measured cost

Task 1, trained seed 11: frozen raw D-Flow versus QGF, IPG, FK and BC

Full-flow endpoint; identity Jacobian only for D-Flow. Each guided condition keeps its own validation-selected settings.



One independently trained model; other trained models remain pending in this snapshot. No test result selects or changes a policy.

Cost includes padded B20 execution and unequal early stopping, particle/flow counts and compilation effects; it is not a Q/NFE query count.

A20 is inherited evidence; B20 is new. Q-averaged identity methods use a separate fresh panel and are not represented by zero bars here.

This preserved figure shows Task1 training seed11 only: ten frozen policies on its original A20 plus fresh B20. The right panel times only newly executed B20 batches. It does not compare the new averaged-Q panel, which has separate starts and may still be pending.

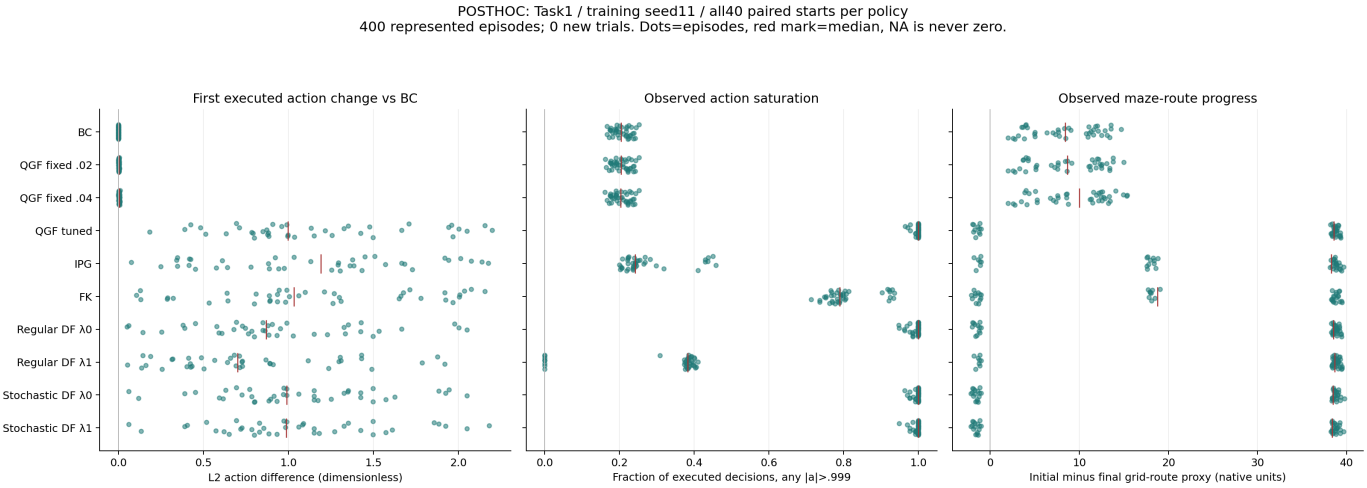
Cost is measured once per batch. It includes unequal inference budgets and early stopping, and can include compilation effects. The three extra successes for regular λ1 relative to tuned QGF in this model do not establish a universal winner or isolate the effect of λ, since the selected iteration counts also differ.

Posthoc saved-trajectory description - Task 1 / model 11

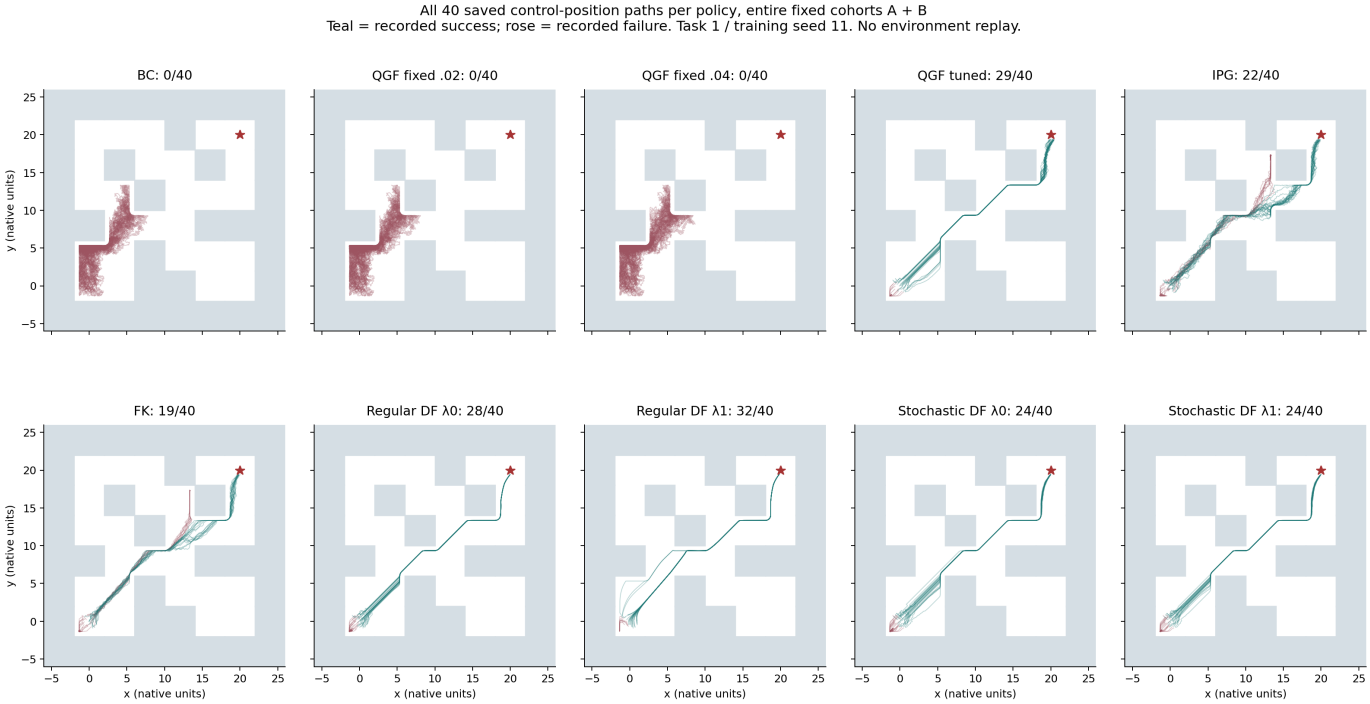
This analysis was specified after outcomes existed. It describes all 400 already represented policy/episode rows from one trained model: 200 inherited A rows and 200 previously executed B rows. It adds 0 trials, model queries or environment steps, and did not select or retune any configuration.

Method	Success / 40	Verified first-action pairs	Missing chosen-action Q rows
BC flow	0	40/40	0
QGF fixed .02	0	40/40	0
QGF fixed .04	0	40/40	0
QGF tuned	29	40/40	0
IPG	22	40/40	40
FK corrector	19	40/40	40
DF Regular, raw Q, λ0	28	40/40	0
DF Regular, raw Q, λ1	32	40/40	0
DF Stochastic, raw Q, λ0	24	40/40	0
DF Stochastic, raw Q, λ1	24	40/40	0

Pairing requires matching initial observation and full native-state hashes, reset/policy/step context and saved source0. Particle0 equality does not match the whole particle ensemble or its selected particle. Own settings and 32 versus 8 flow slices remain confounds. Chosen-action Q is shown only when the exact field was saved; missing is NA. Learned Q is not true value or calibration.



All saved control-position paths - posthoc Task 1 / model 11



All 40 saved paths per policy are shown, including failures. These connect executed control-decision positions rather than MuJoCo internal substeps. The route measure is a cell/BFS proxy, not an exact continuous geodesic; near-zero displacement is not a calibrated collision test. These descriptive outcomes do not isolate a causal noise, prior or averaging effect.

Main paired comparison: Task 1, target 100

Method	Seed 11 observed	Seed 23 observed	Seed 37 observed
BC flow	0/60 partial	0/40 partial	0/40 partial
QGF fixed .02	0/60 partial	0/40 partial	0/40 partial
QGF fixed .04	0/60 partial	0/40 partial	0/40 partial
QGF tuned	47/60 partial	40/40 partial	40/40 partial
IPG	38/60 partial	40/40 partial	38/40 partial
FK corrector	27/60 partial	40/40 partial	40/40 partial
DF Regular, raw Q, λ0	48/60 partial	40/40 partial	40/40 partial
DF Regular, raw Q, λ1	45/60 partial	40/40 partial	40/40 partial
DF Stochastic, raw Q, λ0	46/60 partial	40/40 partial	40/40 partial
DF Stochastic, raw Q, λ1	45/60 partial	40/40 partial	40/40 partial
DF Regular, average Q, λ0	46/60 partial	40/40 partial	40/40 partial
DF Regular, average Q, λ1	48/60 partial	40/40 partial	40/40 partial
DF Stochastic, average Q, λ0	47/60 partial	40/40 partial	40/40 partial
DF Stochastic, average Q, λ1	46/60 partial	40/40 partial	40/40 partial

All cells have target 100. Each shown fraction uses only completed, explicitly recorded B20 request evidence; a missing or partial cell is not 0/100. The 84 primary cells cover 14 methods × 3 trained models × 2 tasks. Required positive-only requests are reported separately.

The original fresh A20+B20 remain unchanged references; C20,D20,E20 extend the same selected configurations without retuning. All 33 guided choices across three models freeze before the task's first main heldout batch. All three models finish every preceding round before a later round release. Paired methods use the same task reset IDs/policy root; disjoint reset IDs give distinct episode-key contexts, without a pooled-independence guarantee.

The historical 612xxx original/extension 40-episode panel above is a different comparison and is never added to these 100. Display aliases consume no additional physical episode when they reference an existing configID. Actual separate numeric-policy reruns still consume cost.

Main paired comparison: Task 5, target 100

Method	Seed 11 observed	Seed 23 observed	Seed 37 observed
BC flow	Pending	Pending	Pending
QGF fixed .02	Pending	Pending	Pending
QGF fixed .04	Pending	Pending	Pending
QGF tuned	Pending	Pending	Pending
IPG	Pending	Pending	Pending
FK corrector	Pending	Pending	Pending
DF Regular, raw Q, λ0	Pending	Pending	Pending
DF Regular, raw Q, λ1	Pending	Pending	Pending
DF Stochastic, raw Q, λ0	Pending	Pending	Pending
DF Stochastic, raw Q, λ1	Pending	Pending	Pending
DF Regular, average Q, λ0	Pending	Pending	Pending
DF Regular, average Q, λ1	Pending	Pending	Pending
DF Stochastic, average Q, λ0	Pending	Pending	Pending
DF Stochastic, average Q, λ1	Pending	Pending	Pending

All cells have target 100. Each shown fraction uses only completed, explicitly recorded B20 request evidence; a missing or partial cell is not 0/100. The 84 primary cells cover 14 methods × 3 trained models × 2 tasks. Required positive-only requests are reported separately.

The original fresh A20+B20 remain unchanged references; C20,D20,E20 extend the same selected configurations without retuning. All 33 guided choices across three models freeze before the task's first main heldout batch. All three models finish every preceding round before a later round release. Paired methods use the same task reset IDs/policy root; disjoint reset IDs give distinct episode-key contexts, without a pooled-independence guarantee.

The historical 612xxx original/extension 40-episode panel above is a different comparison and is never added to these 100. Display aliases consume no additional physical episode when they reference an existing configID. Actual separate numeric-policy reruns still consume cost.

Main target100: completed evidence and incomplete work

Evidence/accounting unit	Observed
Main A/B completed physical episode evidence	1680
Additional C/D/E completed physical episode evidence	280
Main canonical completed B20 CSV batches	98
Main requested display/cohort rows	98
Main required-positive-secondary declarations	0
Observed active/interrupted resume paths	0
Unrequested yet finalized canonical CSVs	0

Across this report’s preserved 40-episode and main 100-episode inputs, each actual physical CSV is counted once: completed new-batch sampler time 4771.479s, completed new-batch wall time 4939.997s. These cumulative batch timers can include compilation and are repeated across episode rows in raw CSVs; they are never multiplied by 20. Unknown timers and unfinished prefix compute are not guessed.

A canonical CSV may be finalized just before its requested display proof is appended. Such real physical evidence and cost are retained separately; a requested result is shown only after its recorded request exists. Volatile .resume.pkl presence is a point-in-time metadata observation, with unknown partial episode/physics/cost counters. No pickle or trajectory arrays are loaded by this report consumer.

Selected settings (1-21 of 84)

Task / seed	Requested condition	κ	η / l	λ	σ / K	N / M	noise / ridge
1 / 11	BC flow	0	-	-	-	1 / 8	-
1 / 11	QGF fixed .02	0.02	-	-	-	1 / 8	-
1 / 11	QGF fixed .04	0.04	-	-	-	1 / 8	-
1 / 11	QGF tuned	80	-	-	-	1 / 8	-
1 / 11	IPG	80	-	-	-	8 / 32	0.005 / 0.1
1 / 11	FK corrector	80	-	-	-	16 / 32	0.02 / -
1 / 11	DF Regular, raw Q, λ0	5	0.1 / 256	0	-	1 / 8	-
1 / 11	DF Regular, raw Q, λ1	5	0.1 / 128	1	-	1 / 8	-
1 / 11	DF Stochastic, raw Q, λ0	80	0.01 / 128	0	-	1 / 8	-
1 / 11	DF Stochastic, raw Q, λ1	80	0.01 / 128	1	-	1 / 8	-
1 / 11	DF Regular, average Q, λ0	20	0.1 / 256	0	0.1 / 16	1 / 8	-
1 / 11	DF Regular, average Q, λ1	80	0.03 / 256	1	0.03 / 16	1 / 8	-
1 / 11	DF Stochastic, average Q, λ0	80	0.03 / 128	0	0.1 / 4	1 / 8	-
1 / 11	DF Stochastic, average Q, λ1	80	0.03 / 256	1	0.2 / 4	1 / 8	-
1 / 23	BC flow	0	-	-	-	1 / 8	-
1 / 23	QGF fixed .02	0.02	-	-	-	1 / 8	-
1 / 23	QGF fixed .04	0.04	-	-	-	1 / 8	-
1 / 23	QGF tuned	80	-	-	-	1 / 8	-
1 / 23	IPG	40	-	-	-	16 / 32	0.02 / 0.01
1 / 23	FK corrector	80	-	-	-	16 / 32	0.02 / -
1 / 23	DF Regular, raw Q, λ0	40	0.1 / 256	0	-	1 / 8	-

N=particles; M=flow time slices. A dash means unused or still pending, not zero. Stochastic D-Flow uses T=1; particle noise is a separate setting. FK does not use IPG ridge. Complete configuration dictionaries and selection states are in SELECTED_SETTINGS.json. Individually selected new-Qavg settings may still await the all-three-model test barrier.

Selected settings (22-42 of 84)

Task / seed	Requested condition	κ	η / I	λ	σ / K	N / M	noise / ridge
1 / 23	DF Regular, raw Q, λ1	80	0.03 / 256	1	-	1 / 8	-
1 / 23	DF Stochastic, raw Q, λ0	80	0.03 / 256	0	-	1 / 8	-
1 / 23	DF Stochastic, raw Q, λ1	80	0.03 / 128	1	-	1 / 8	-
1 / 23	DF Regular, average Q, λ0	40	0.1 / 256	0	0.2 / 16	1 / 8	-
1 / 23	DF Regular, average Q, λ1	80	0.03 / 256	1	0.2 / 4	1 / 8	-
1 / 23	DF Stochastic, average Q, λ0	80	0.1 / 256	0	0.03 / 16	1 / 8	-
1 / 23	DF Stochastic, average Q, λ1	80	0.03 / 128	1	0.2 / 4	1 / 8	-
1 / 37	BC flow	0	-	-	-	1 / 8	-
1 / 37	QGF fixed .02	0.02	-	-	-	1 / 8	-
1 / 37	QGF fixed .04	0.04	-	-	-	1 / 8	-
1 / 37	QGF tuned	40	-	-	-	1 / 8	-
1 / 37	IPG	40	-	-	-	16 / 8	0.02 / 0.01
1 / 37	FK corrector	80	-	-	-	16 / 32	0.02 / -
1 / 37	DF Regular, raw Q, λ0	5	0.1 / 256	0	-	1 / 8	-
1 / 37	DF Regular, raw Q, λ1	80	0.03 / 256	1	-	1 / 8	-
1 / 37	DF Stochastic, raw Q, λ0	80	0.1 / 256	0	-	1 / 8	-
1 / 37	DF Stochastic, raw Q, λ1	80	0.03 / 128	1	-	1 / 8	-
1 / 37	DF Regular, average Q, λ0	40	0.1 / 128	0	0.2 / 4	1 / 8	-
1 / 37	DF Regular, average Q, λ1	80	0.03 / 256	1	0.2 / 4	1 / 8	-
1 / 37	DF Stochastic, average Q, λ0	80	0.03 / 128	0	0.2 / 4	1 / 8	-
1 / 37	DF Stochastic, average Q, λ1	80	0.03 / 128	1	0.2 / 4	1 / 8	-

N=particles; M=flow time slices. A dash means unused or still pending, not zero. Stochastic D-Flow uses T=1; particle noise is a separate setting. FK does not use IPG ridge. Complete configuration dictionaries and selection states are in SELECTED_SETTINGS.json. Individually selected new-Qavg settings may still await the all-three-model test barrier.

Selected settings (43-63 of 84)

Task / seed	Requested condition	κ	η / l	λ	σ / K	N / M	noise / ridge
5 / 11	BC flow	0	-	-	-	1 / 8	-
5 / 11	QGF fixed .02	0.02	-	-	-	1 / 8	-
5 / 11	QGF fixed .04	0.04	-	-	-	1 / 8	-
5 / 11	QGF tuned	20	-	-	-	1 / 8	-
5 / 11	IPG	20	-	-	-	16 / 32	0.005 / 0.01
5 / 11	FK corrector	80	-	-	-	8 / 32	0.02 / -
5 / 11	DF Regular, raw Q, λ0	1	0.1 / 256	0	-	1 / 8	-
5 / 11	DF Regular, raw Q, λ1	5	0.1 / 128	1	-	1 / 8	-
5 / 11	DF Stochastic, raw Q, λ0	80	0.1 / 256	0	-	1 / 8	-
5 / 11	DF Stochastic, raw Q, λ1	5	0.03 / 128	1	-	1 / 8	-
5 / 11	DF Regular, average Q, λ0	1	0.1 / 256	0	0.01 / 16	1 / 8	-
5 / 11	DF Regular, average Q, λ1	5	0.1 / 256	1	0.01 / 16	1 / 8	-
5 / 11	DF Stochastic, average Q, λ0	40	0.005 / 256	0	0.01 / 4	1 / 8	-
5 / 11	DF Stochastic, average Q, λ1	80	0.005 / 256	1	0.01 / 16	1 / 8	-
5 / 23	BC flow	0	-	-	-	1 / 8	-
5 / 23	QGF fixed .02	0.02	-	-	-	1 / 8	-
5 / 23	QGF fixed .04	0.04	-	-	-	1 / 8	-
5 / 23	QGF tuned	40	-	-	-	1 / 8	-
5 / 23	IPG	5	-	-	-	8 / 8	0 / 0.01
5 / 23	FK corrector	5	-	-	-	16 / 32	0 / -
5 / 23	DF Regular, raw Q, λ0	5	0.1 / 128	0	-	1 / 8	-

N=particles; M=flow time slices. A dash means unused or still pending, not zero. Stochastic D-Flow uses T=1; particle noise is a separate setting. FK does not use IPG ridge. Complete configuration dictionaries and selection states are in SELECTED_SETTINGS.json. Individually selected new-Qavg settings may still await the all-three-model test barrier.

Selected settings (64-84 of 84)

Task / seed	Requested condition	κ	η / l	λ	σ / K	N / M	noise / ridge
5 / 23	DF Regular, raw Q, λ1	5	0.1 / 128	1	-	1 / 8	-
5 / 23	DF Stochastic, raw Q, λ0	5	0.1 / 128	0	-	1 / 8	-
5 / 23	DF Stochastic, raw Q, λ1	40	0.01 / 2	1	-	1 / 8	-
5 / 23	DF Regular, average Q, λ0	80	0.1 / 8	0	0.01 / 16	1 / 8	-
5 / 23	DF Regular, average Q, λ1 (pending)	-	-	-	-	-	-
5 / 23	DF Stochastic, average Q, λ0 (pending)	-	-	-	-	-	-
5 / 23	DF Stochastic, average Q, λ1 (pending)	-	-	-	-	-	-
5 / 37	BC flow	0	-	-	-	1 / 8	-
5 / 37	QGF fixed .02	0.02	-	-	-	1 / 8	-
5 / 37	QGF fixed .04	0.04	-	-	-	1 / 8	-
5 / 37	QGF tuned	40	-	-	-	1 / 8	-
5 / 37	IPG	8	-	-	-	8 / 32	0 / 0.01
5 / 37	FK corrector	40	-	-	-	16 / 8	0.02 / -
5 / 37	DF Regular, raw Q, λ0	5	0.1 / 128	0	-	1 / 8	-
5 / 37	DF Regular, raw Q, λ1	80	0.1 / 16	1	-	1 / 8	-
5 / 37	DF Stochastic, raw Q, λ0	80	0.1 / 8	0	-	1 / 8	-
5 / 37	DF Stochastic, raw Q, λ1	80	0.1 / 8	1	-	1 / 8	-
5 / 37	DF Regular, average Q, λ0 (pending)	-	-	-	-	-	-
5 / 37	DF Regular, average Q, λ1 (pending)	-	-	-	-	-	-
5 / 37	DF Stochastic, average Q, λ0 (pending)	-	-	-	-	-	-
5 / 37	DF Stochastic, average Q, λ1 (pending)	-	-	-	-	-	-

N=particles; M=flow time slices. A dash means unused or still pending, not zero. Stochastic D-Flow uses T=1; particle noise is a separate setting. FK does not use IPG ridge. Complete configuration dictionaries and selection states are in SELECTED_SETTINGS.json. Individually selected new-Qavg settings may still await the all-three-model test barrier.

Failures, limits and numerical checks

All six model math/profile gates completed. The preserved raw float32 finite-difference checks include 6 failures out of 774 raw rows. Separately declared reference criteria passed: 678 non-FD checks and 96 double-reference comparisons, with 0 qualified failures. No tolerance, checkpoint, production precision or numerical-source change was made to erase the raw failures.

Engineering profile: 84 settings, 168 offline sampler calls and 3360 action rows. These are not physical episodes. The largest observed steady B20 call was 0.371 seconds; this is not a full-episode runtime guarantee.

A source-description error was found and corrected before fresh40 release: actual reset IDs, not row indices 0-19, enter the source-key folding. Disjoint A/B reset IDs create 40 distinct episode-key contexts under a shared policy root. The old wording is preserved; numerical code and cohorts were unchanged.

A repeated canonical BC validation control can consume real compute while reproducing already observed trajectories. Such repeats count as executions/cost but not new independent trajectory evidence. Raw completed-record counters after a resume may include an earlier partial prefix; report unique completed evidence or subtract separately preserved prefix work before claiming incremental attempt cost.

Q averaging may suppress a narrow false peak or a genuine narrow optimum. Noise may help movement within the actor’s image or amplify a mistaken score. Native success alone does not identify which mechanism occurred. Learned-V Bellman residuals measure self-consistency, not ground-truth Q error. Saved traces do not contain every latent, perturbation bank or counterfactual Q value.

Source-only boundary note (0 new queries): the current positive-σ objective differentiates through clipping of perturbed actions, even though identity guidance bypasses the separate flow-endpoint Jacobian. For Q(a)=a at a=1 with offsets ±δ, 0<δ<2, the averaged score derivative is1/2 versus the raw derivative1. In multiple dimensions clip masks can also change direction; small positive σ need not recover a boundary gradient. This mathematical property is not a causal explanation of these outcomes. A prospective fixed raw/Qavg strength control remains unimplemented and unrun; no current method, setting or comparison changes.

Small validation panels and hierarchical search can miss useful settings or select noisy winners. Unequal inference iterations, particle/flow counts, early stopping and compilation confound runtime comparisons. One trained model cannot establish a universal method winner. All completed zeros remain in the result tables; pending and interrupted cells remain explicit.

The auxiliary panels below are separate diagnostic validation gates. They are not Task2 full-domain tuning, a new D-Flow comparison, or any part of the main Task1/5 target100 heldout sample.

Auxiliary diagnostic: Task 2 competence

Status: COMPLETE_AUDITED. Diagnostic validation only; separate from all main heldout denominators. No training, new tuning or algorithm ranking.

Official medium-maze Task2 starts around (0,20) and aims for (20,0), with the original1000-control cap. The gate uses the exact frozen BC1k and Task2-specific IQL Q/V50k for models11/23/37. Four locked policies on8 fresh diagnostic starts per model request96 executions.

The tuned reference is locked historical validation-tuned QGF: κ12/40/20 for models11/23/37. It is not a new current-domain sweep including κ80, and does not establish future full-comparison QGF selection as complete. Older historical60/60 is readiness context, not a harder-task or future100 guarantee.

Diagnostic condition	Seed11	Seed23	Seed37
BC	0/8	0/8	0/8
QGF fixed .02	0/8	0/8	0/8
QGF fixed .04	0/8	0/8	0/8
QGF historical VAL-tuned	8/8	8/8	7/8

Observed completed diagnostic validation executions: 96 of96 requested. Invalid action/native steps: 0; nonfinite-return episodes: 0. All zeros and failures are retained; unfinished physical prefix cost is unknown.

Saved-data audit:
outputs/overnight_research_20261106_night/review/task2_competence_completed/20261107T070757_505892Z/REVIEW.json
SHA256 231ffe107c2dd5924b4d7b42f30a4f470c4f0ca774e35fa3c257eaca5d0f2d9f

Auxiliary diagnostic: local coverage topology

Status: COMPLETE_AUDITED. Diagnostic validation only; separate from all main heldout denominators. No training, new tuning or algorithm ranking.

This separate stress diagnostic changes the reset distribution to fixed logged-TRAIN positions around the coverage region and uses a128-control cap. Upstream/downstream strata are constructed and frozen before model queries; goal/topology follow the declared native maze. It is not the standard1000-control benchmark.

Three trained models each supply FULL/HOLE critic arms. The hole withholds an action cone only from Q/V training; both arms keep the same full-data BC actor. BC and fixedκ40 QGF are evaluated on4 starts per stratum, two strata and two arms:96 actual requests. BC arm repeats and paired starts do not add independently trained models. No true-Q calibration or causal coverage gain is established.

Diagnostic condition	Seed11	Seed23	Seed37
upstream / FULL / BC	0/4	0/4	0/4
upstream / FULL / QGF fixed40	4/4	2/4	4/4
upstream / HOLE / BC	0/4	0/4	0/4
upstream / HOLE / QGF fixed40	4/4	4/4	4/4
downstream / FULL / BC	0/4	0/4	0/4
downstream / FULL / QGF fixed40	4/4	4/4	4/4
downstream / HOLE / BC	0/4	0/4	0/4
downstream / HOLE / QGF fixed40	4/4	4/4	4/4

Observed completed diagnostic validation executions: 96 of96 requested. Invalid action/native steps: 0; nonfinite-return episodes: 0. All zeros and failures are retained; unfinished physical prefix cost is unknown.

Saved-data audit:
outputs/overnight_research_20261106_night/review/coverage_topology_gate_completed/20261107T072144_992144Z/REVIEW.json
SHA256 13030dd477c668a3369a50621c150ba4f481fe8b0d0f378a3903fd979b7a1410

Expanded QGF validation: separate FULL / HOLE follow-up

Status: COMPLETE_AUDITED. This is a separate validation-only follow-up on newly reserved, logged-TRAIN reset positions with a128-control cap. It is not the standard Task1/5 heldout target100 panel and adds no heldout evidence to its denominators.

The independently declared26-strength grid is κ=0,.02,.04,.06,.08,.1,.2,.3,.4,.6,.8,1,2,3,5,8,10,12,16,20,32,40,64,80,160,320. Every strength receives the same balanced8 validation starts:4 upstream and4 downstream, independently for each of six critic-arm/model cells. Complete execution is156 B8 batches /1,248 actual validation trials. κ0 is the canonical BC behavior for this QGF kernel.

Model / critic arm	Own κ	VAL /8	Up /4	Down /4	Positive κ
11 / FULL	64	8/8	4/4	4/4	64
11 / HOLE	40	8/8	4/4	4/4	40
23 / FULL	32	6/8	2/4	4/4	32
23 / HOLE	40	8/8	4/4	4/4	40
37 / FULL	80	8/8	4/4	4/4	80
37 / HOLE	40	8/8	4/4	4/4	40

Observed completed validation: 156 B8 batches / 1248 actual episodes. Completed zero-success batches: 68; completed invalid action steps: 0; completed nonfinite-return episodes: 0. All candidate outcomes remain in the bound raw evidence, including zeros and failures; unfinished prefix outcomes and cost are unknown.

Saved-data audit:
outputs/overnight_research_20261106_night/review/tuned_topology_qgf26_completed/20261107T075453_232660Z/REVIEW.json
SHA256 22f29f3a3847a9cafb8b3451a4336ac4522a30fd875745b29c20910ecc2417d1

Three trained BC actors are shared across FULL/HOLE critic arms. The action-cone hole affects only Q/V training; both arms retain the full-data BC actor. Each arm chooses its own validation setting, so differences combine critic coverage and tuning. Validation success does not establish true-Q calibration, a heldout winner or a causal benefit from deleting data.

The proposed D-Flow extension retains all eight identity/full-flow conditions with independently fixed λ0/1, expanded I through512 and Qavg strata. Its actual512 engineering/profiles, D-Flow/IPG/FK validation, all66-guided-choice global freeze and future100 tests remain deferred. The six-QGF freeze is explicitly not that full-comparison freeze; this appendix authorizes no further queries.

Reproducibility and preserved campaigns

The evidence JSON binds complete CSV/metadata/trace triples with SHA256, exact reset IDs, policy roots, model inputs and selected configurations. New physical evidence and inherited references are counted separately. Selection uses validation only; heldout outcomes never retune or choose a family configuration.

Native model construction combines the frozen BC policy with the frozen goal-specific IQL critic. The two-head target minimum is used for guidance. Shared model hashes and full-agent tree hashes are checked by released workers before/after execution. Local CPU only: at most two scientific workers, bounded 7200-second attempts, explicit reviewed checkpoint resumes, no automatic failure retries.

Task / seed	BC checkpoint SHA prefix	IQL checkpoint SHA prefix
1 / 11	45496ed77dd755a7	9bb2d24f10135780
1 / 23	ce24a7f3c86bbfaa	e5e09273f762357a
1 / 37	1a01ebdb0ed1d8a9	995c4299ccabef98
5 / 11	45496ed77dd755a7	fc61e43f2794c8c4
5 / 23	ce24a7f3c86bbfaa	d07c86d71ed81012
5 / 37	1a01ebdb0ed1d8a9	d6ec49560a722a9f

Verified common configuration: H=1, flow slices=8, two Q heads with minimum aggregation; discount=0.99, expectile=0.8, target τ=0.005, batch=256, learning rates=0.0003. Actor hidden layers=[64, 64]; Q/V hidden layers=[128, 128]. Full checkpoint paths, configuration dictionaries and SHA256 values are included in the bound manifest/reference bundle.

Core numerical source hashes

work/overnight_research_20261105/samplers.py
a615c61d284cb256e1291a5ae71cc719b03614a9218f84a49e86dfa6a80105e9

work/overnight_research_20261106_night/identity_math/kernel.py
0f8d7a4519ba95cc3e8aafcfc3996e381408c9dbc44ac555cb0ccae3cab79d56

work/overnight_research_20261106_continuation/exploratory_identity_qavg_v2/kernel.py
7d7d10799f5f3b81aad15668568c477aaf95bffe35949c47b701ec753ba01ddc

Historical reports and archives remain separate. The earlier all-experiments report includes synthetic one-point diagnostics, custom sequential learned-BC/IQL navigation, modified FULL/HOLE coverage studies and MuJoCo cube manipulation. FULL/HOLE labels concern older offline-data coverage arms; they do not mean full-flow versus Euler endpoints here.

Final archive verification: pending the final morning integrity receipt; this preview is not final delivery.

A next discriminating experiment should be preregistered after this study and use new data: match inference budgets, repeat trained-model seeds, and separate local narrow-peak errors from broad critic errors using a controlled reference. No such new experiment or retuning is performed from these heldout results.